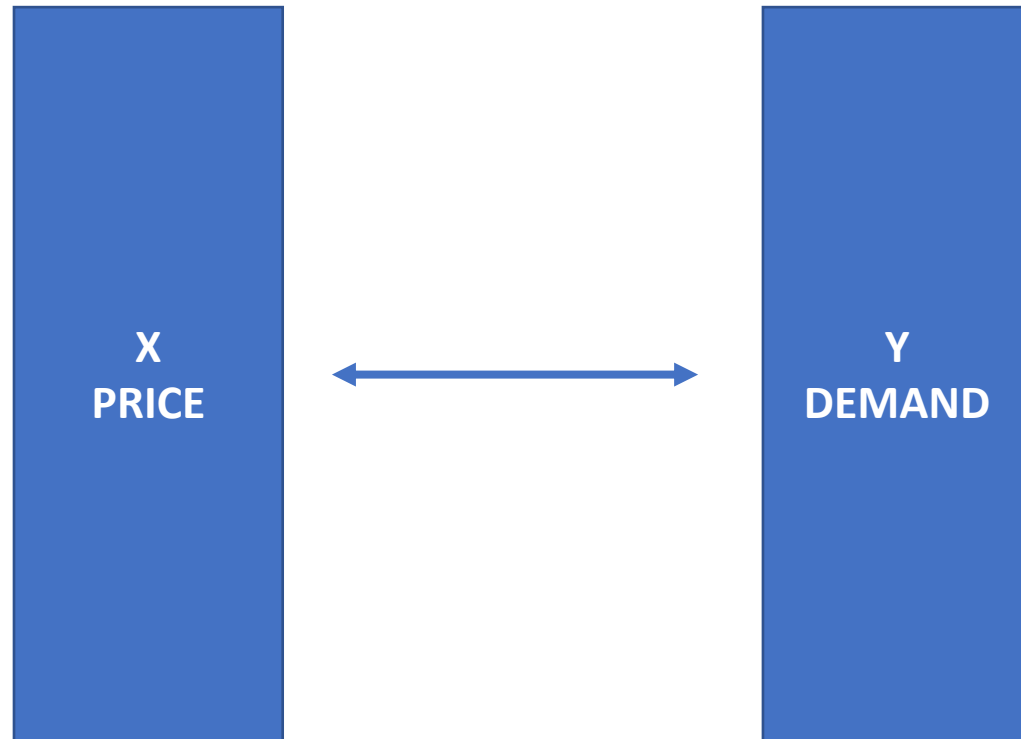


Measures of association:

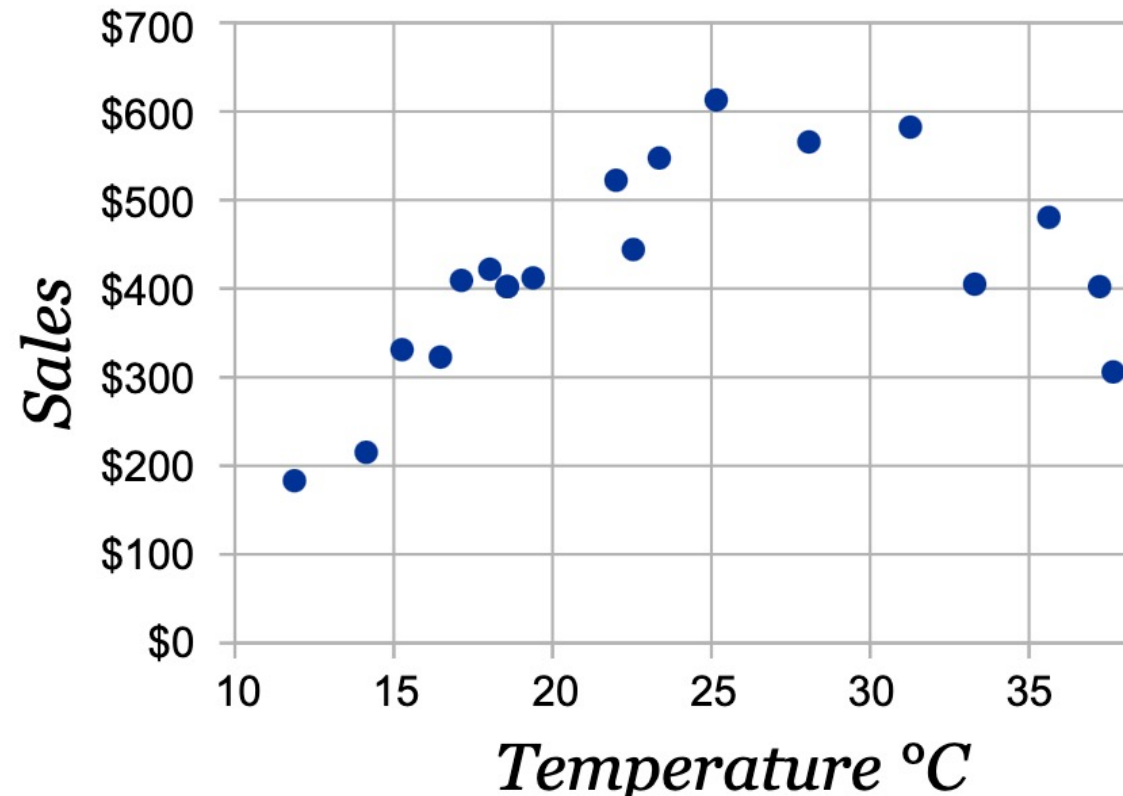
In statistics, any of various factors or coefficients used to quantify a relationship between two or more variables.



Bivariate Data

Data for **two variables** (usually two types of related data).

Example: Ice cream sales versus the temperature on that day. The two variables are Ice Cream Sales and Temperature.



Example:

For example, a researcher may record how long it takes people to complete a crossword puzzle while measuring the stress levels of the participants. In this example, **the two variables that the researcher is examining are time and stress.**

Bivariate Relationship

One of the main purposes of determining whether or not two variables are related is to see if **one variable causes the other**. This type of research involves two basic types of variables:

Independent variable: The variable that a researcher manipulates in an experiment. The researcher predicts that changing the independent variable causes a meaningful change in another variable.

Dependent variable: The variable that a researcher predicts will change by manipulating the independent variable. The important thing to note is that the researcher does not directly change this variable; it is changed or controlled by an outside factor (often the independent variable).

.

Example:

Hours Studied	Exam Grade
4	76
0	61
1	71
8	82
11	97

Covariance

covariance is a measure of the relationship between two random variables. The metric evaluates **how much – to what extent – the variables change together**. In other words, it is essentially a measure of the variance between two variables.

covariance is measured in units.

The units are computed by multiplying the units of the two variables. The variance can take any positive or negative values. The values are interpreted as follows:

Types of covariance

- **Positive covariance:** Indicates that two variables tend to **move in the same direction**.
- **Negative covariance:** Reveals that two variables tend to **move in inverse directions**.

$$\text{Cov} (X, Y) = \frac{\sum (X_i - \bar{X})(Y_j - \bar{Y})}{n - 1}$$

Where:

- X_i – the values of the X-variable
- Y_j – the values of the Y-variable
- $\bar{X}$ – the mean (average) of the X-variable
- $\bar{Y}$ – the mean (average) of the Y-variable
- n – the number of data points

	S&P 500	ABC Corp.
2013	1,692	68
2014	1,978	102
2015	1,884	110
2016	2,151	112
2017	2,519	154

	S&P 500	ABC Corp.	a	b	a x b
2013	1,692	68	-352.80	-41.20	14,535.36
2014	1,978	102	-66.80	-7.20	480.96
2015	1,884	110	-160.80	0.80	-128.64
2016	2,151	112	106.20	2.80	297.36
2017	2,519	154	474.20	44.80	21,244.16
Mean	2,044.80	109.20	Sum		36,429.20

$$\text{Cov (X, Y)} = \frac{\sum (X_i - \bar{X})(Y_j - \bar{Y})}{n - 1}$$

$$\text{Mean (S\&P 500)} = \frac{1,692 + 1,978 + 1,884 + 2,151 + 2,519}{5} = 2,044.80$$

$$\text{Mean (ABC Corp.)} = \frac{68 + 102 + 110 + 112 + 154}{5} = 109.20$$

$$\text{Cov(S\&P 500, ABC Corp.)} = \frac{36,429.20}{5-1} = 9,107.30$$

Correlation

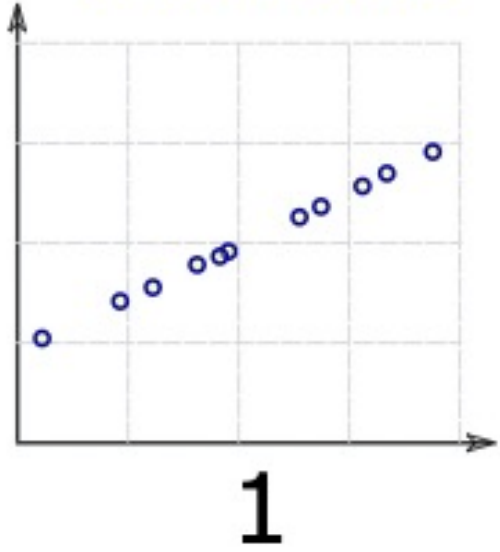
When two sets of data are strongly linked together we say they have a **High Correlation**.

The word Correlation is made of **Co-** (meaning "together"), and **Relation**

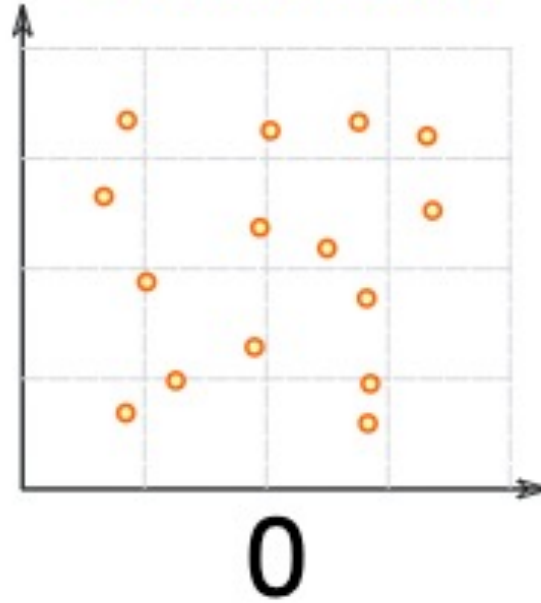
Correlation is **Positive** when the values **increase** together, and

Correlation is **Negative** when one value **decreases** as the other increases

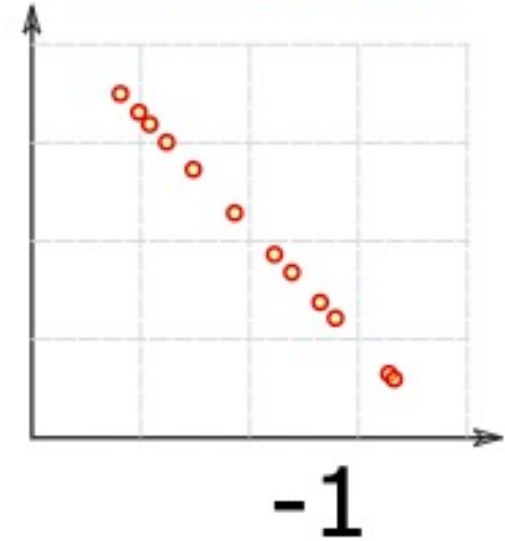
*Perfect
Positive
Correlation*



*No
Correlation*



*Perfect
Negative
Correlation*



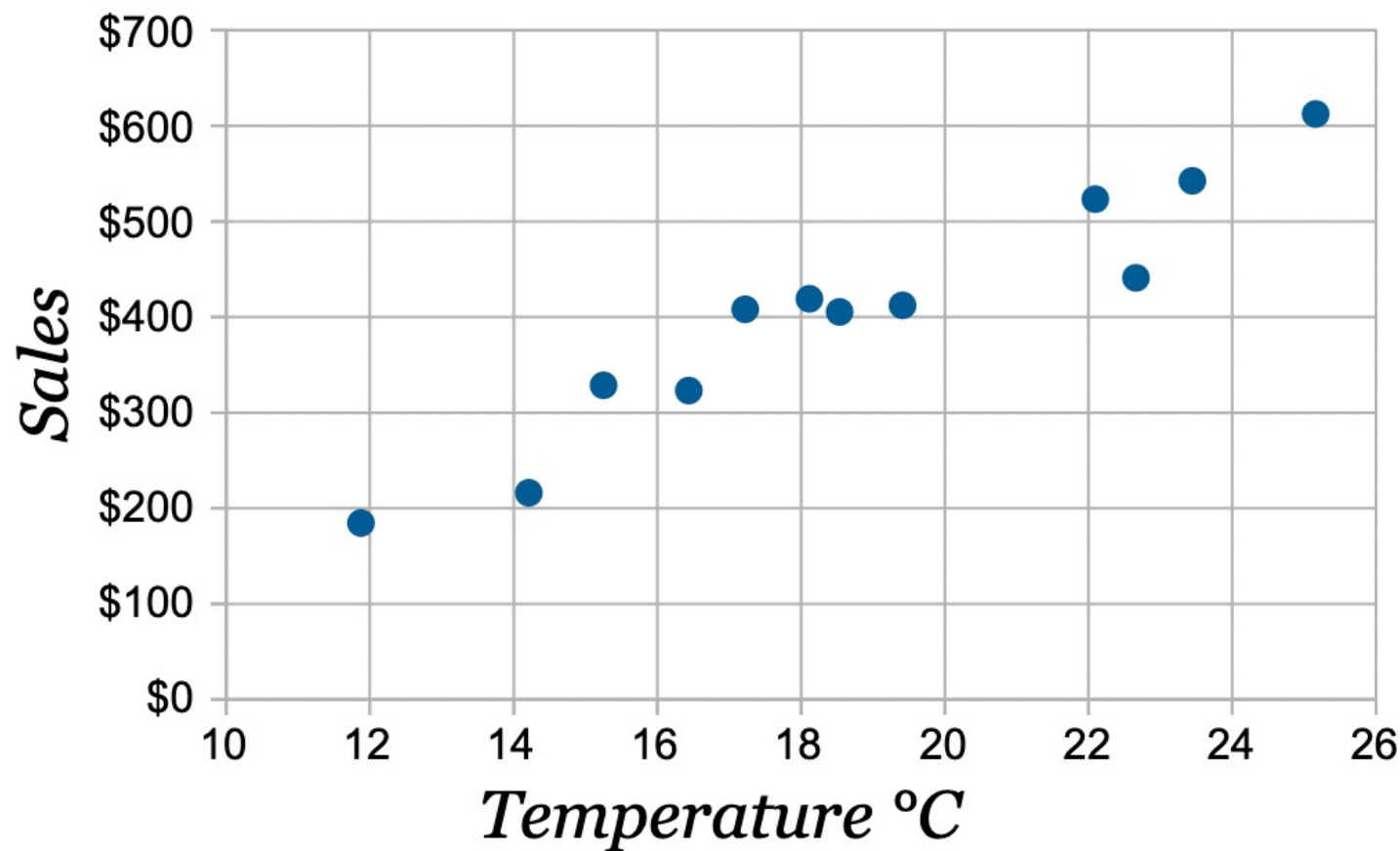
Correlation can have a value:

- **1** is a perfect positive correlation
- **0** is no correlation (the values don't seem linked at all)
- **-1** is a perfect negative correlation

Example: Ice Cream Sales

The local ice cream shop keeps track of how much ice cream they sell versus the temperature on that day. Here are their figures for the last 12 days:

Ice Cream Sales vs Temperature	
Temperature °C	Ice Cream Sales
14.2°	\$215
16.4°	\$325
11.9°	\$185
15.2°	\$332
18.5°	\$406
22.1°	\$522
19.4°	\$412
25.1°	\$614
23.4°	\$544
18.1°	\$421
22.6°	\$445
17.2°	\$408



Calculating the Correlation

Now we're ready to compute the correlation value. The formula for the correlation is:

$$r = \frac{N \sum xy - \sum x \sum y}{\sqrt{(N \sum x^2 - (\sum x)^2)(N \sum y^2 - (\sum y)^2)}}$$

where:

- N is the number of pairs of scores,
- $\sum xy$ is the sum of the products of paired scores,
- $\sum x$ is the sum of x scores,
- $\sum y$ is the sum of y scores,
- $\sum x^2$ is the sum of squared x scores,
- $\sum y^2$ is the sum of squared y scores.

Person	Height (x)	Self Esteem (y)	x*y	x*x	y*y
1	68	4.1	278.8	4624	16.81
2	71	4.6	326.6	5041	21.16
3	62	3.8	235.6	3844	14.44
4	75	4.4	330	5625	19.36
5	58	3.2	185.6	3364	10.24
6	60	3.1	186	3600	9.61
7	67	3.8	254.6	4489	14.44
8	68	4.1	278.8	4624	16.81
9	71	4.3	305.3	5041	18.49
10	69	3.7	255.3	4761	13.69
11	68	3.5	238	4624	12.25

Calculating the Correlation

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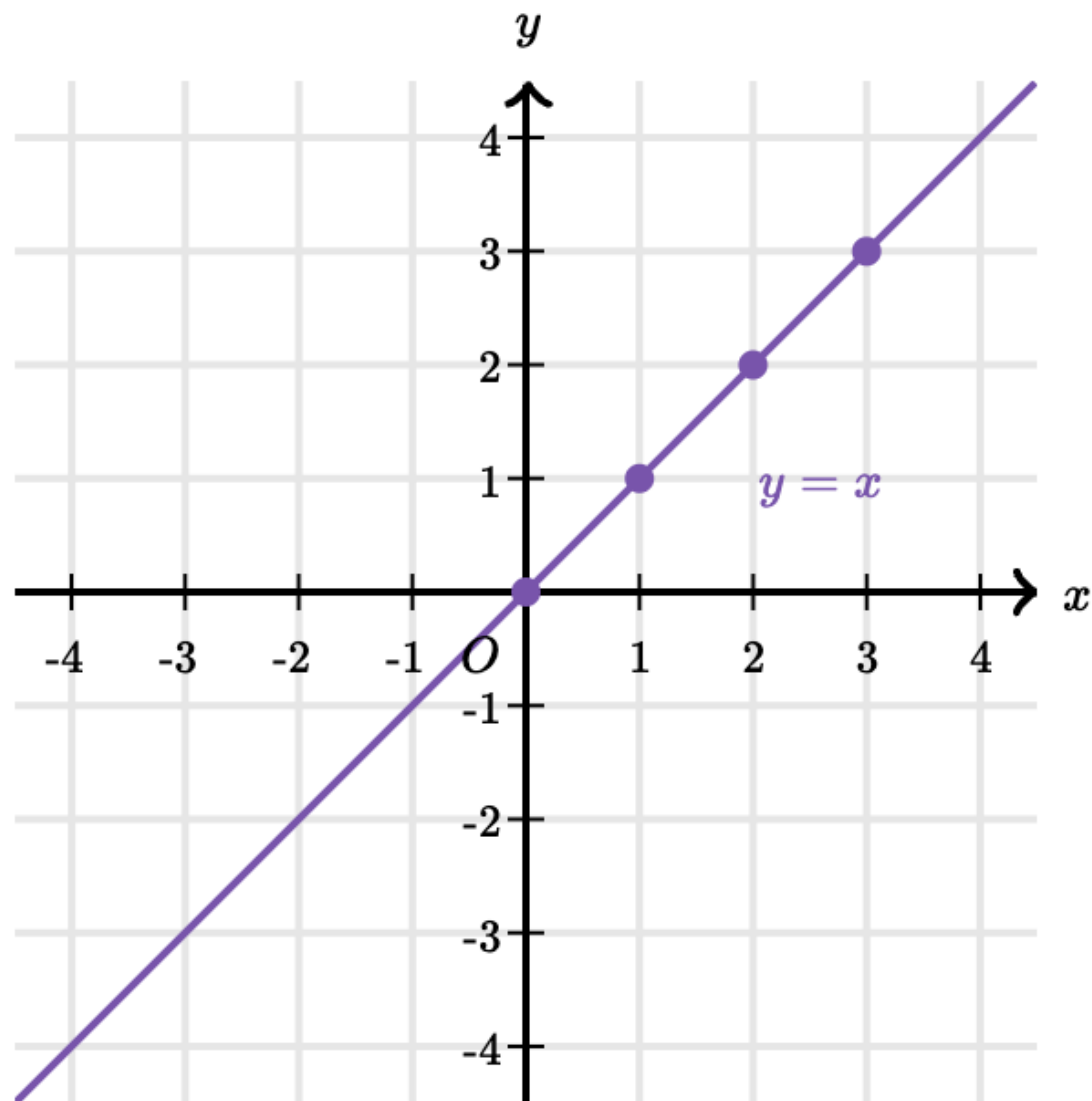
$$\begin{aligned}
r &= \frac{N \sum xy - \sum x \sum y}{\sqrt{(N \sum x^2 - (\sum x)^2)(N \sum y^2 - (\sum y)^2)}} = \\
&= \frac{20 \cdot 4937.6 - 1308 \cdot 75.1}{\sqrt{(20 \cdot 85912 - 1308^2)(20 \cdot 285.45 - 75.1^2)}} = \\
&= \frac{98752 - 98230.8}{\sqrt{(1718240 - 1710864)(5709 - 5640.1)}} = \\
&= \frac{521.2}{\sqrt{7376 \cdot 68.99}} = \frac{521.2}{\sqrt{508870.2}} = \\
&= \frac{521.2}{713.3514} = 0.73
\end{aligned}$$

Spearman correlation

It measures the strength and direction of the association between two ranked variables. But before we talk about the Spearman correlation coefficient, it is important to understand Pearson's correlation first. A Pearson correlation is a statistical measure of the **strength of a linear relationship between paired data**.

Linear relationship

x	y
0	0
1	1
2	2
3	3



Requirements for plotting Spearman correlation

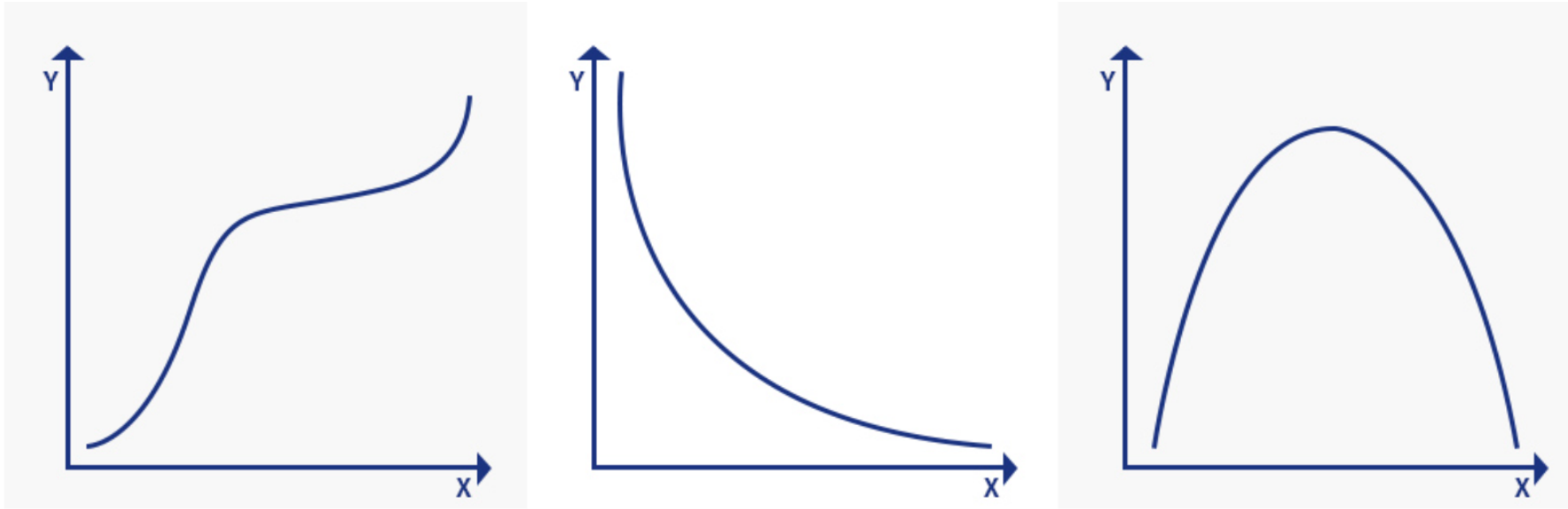
For the calculation and significance testing of the ranking variable, it requires the following data assumption to hold true:

- Interval or ratio level
- Linearly related
- Bivariant distributed

If requirements doesn't meet ! then we need to work with Spearman's rank coefficient of correlation Named after Charles Spearman

Spearman's rank coefficient of correlation

It is necessary to know what **monotonic function** is to understand Spearman correlation coefficient. A **monotonic function** is one that either never decreases or never increases as it is an independent variable increase.



The image explains three concepts in monotonic function:

1. Monotonically increasing: When the 'x' variable increases and the 'y' variable never decreases.
2. Monotonically decreasing: When the 'x' variable increases but the 'y' variable never increases
3. Not monotonic: When the 'x' variable increases and the 'y' variable sometimes increases and sometimes decreases.